

OE-EC604A — Electronic Measurement & Measuring Instruments

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Module 5 Exam Notes: Bridges, Parameter Measurement & DAS

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Contents

1	DC and AC Bridges	2
1.1	Wheatstone Bridge	2
1.1.1	Derivation of Balance Equation	2
1.1.2	Limitations of Wheatstone Bridge	2
1.2	Kelvin Double Bridge	3
1.2.1	Derivation of Balance Equation	3
1.3	Maxwell Inductance-Capacitance Bridge	4
1.3.1	Derivation of Balance Equation	4
1.3.2	Advantages and Disadvantages	5
2	Measurement of Non-Electrical Parameters	6
2.1	Flow Measurement: Electromagnetic Flow Meter	6
2.1.1	Concept and Working	6
2.2	Level Measurement	6
2.2.1	Capacitive Level Transducer	6
2.2.2	Ultrasonic Level Transducer	6
2.3	Pressure Measurement	7
2.3.1	LVDT-Based Pressure Cell	7
2.3.2	Pirani Vacuum Gauge	7
2.3.3	McLeod Vacuum Gauge	7
3	Data Acquisition Systems (DAS)	8
3.1	Analog and Digital DAS	8
3.2	Roles of Key Digital DAS Components	8

1. DC and AC Bridges

Bridges are comparison circuits used to measure resistance, inductance, capacitance, and impedance with high precision.

1.1 Wheatstone Bridge

The **Wheatstone Bridge** is a fundamental DC bridge used for measuring medium resistances ($1\ \Omega$ to $100\ \text{k}\Omega$).

Wheatstone Bridge Principle

It consists of four resistive arms forming a closed loop, with a DC source connected across one pair of opposite junctions and a null detector (galvanometer) across the other. When the potential difference across the galvanometer is zero, the bridge is balanced.

1.1.1 Derivation of Balance Equation

Let the four resistive arms be P , Q , R , and S :

- P and Q are fixed, known ratio arms.
- R is the variable standard resistance.
- S is the unknown resistance (R_x).

At balance, the current through the galvanometer is $I_g = 0$. This implies:

$$V_B = V_D \implies I_1 P = I_2 R \quad (1)$$

Since no current passes through the galvanometer, the current in arm BC is I_1 and in arm DC is I_2 :

$$I_1 = \frac{E}{P + Q} \quad \text{and} \quad I_2 = \frac{E}{R + S} \quad (2)$$

Substituting these currents into the potential equation:

$$\left(\frac{E}{P + Q}\right) P = \left(\frac{E}{R + S}\right) R \implies \frac{P}{P + Q} = \frac{R}{R + S} \quad (3)$$

Cross-multiplying yields:

$$P(R + S) = R(P + Q) \implies PR + PS = PR + QR \implies PS = QR \implies S = \frac{Q}{P} R \quad (4)$$

Thus, the unknown resistance is:

$$R_x = \frac{Q}{P} R \quad (5)$$

1.1.2 Limitations of Wheatstone Bridge

It cannot measure low resistances ($< 1\ \Omega$) accurately because the contact resistances of terminals and the lead wire resistances are of similar magnitude, introducing significant errors.

1.2 Kelvin Double Bridge

The **Kelvin Double Bridge** is a modification of the Wheatstone bridge designed specifically to eliminate lead and contact resistances when measuring low resistances ($1 \mu\Omega$ to 1Ω).

Kelvin Double Bridge Concept

It introduces a second set of ratio arms (p and q) which are connected to a galvanometer to bypass the potential drop across the low-resistance connecting link (r) between the unknown low resistance (R) and standard low resistance (S).

1.2.1 Derivation of Balance Equation

Let the main ratio arms be P and Q , and the inner ratio arms be p and q . The resistance of the connecting lead is r . The galvanometer is connected between the junction of P and Q and the junction of p and q . At balance, the potential difference across the galvanometer is zero. The current I divides at the connecting link:

- Main branch current is I .
- The path through the inner ratio arms ($p + q$) is in parallel with the link resistance r .

The current through the link branch $p + q$ is:

$$I_{pq} = I \cdot \frac{r}{p + q + r} \quad (6)$$

The voltage drop across the branch p is:

$$V_p = I_{pq} \cdot p = I \left(\frac{pr}{p + q + r} \right) \quad (7)$$

For balance, the ratio of potential drops across the primary arms must be equal:

$$\frac{P}{Q} = \frac{R + V_p}{S + V_q} = \frac{R + I \left(\frac{pr}{p + q + r} \right)}{S + I \left(\frac{qr}{p + q + r} \right)} \quad (8)$$

Cross-multiplying and solving for R :

$$P \left[S + \frac{qr}{p + q + r} \right] = Q \left[R + \frac{pr}{p + q + r} \right] \quad (9)$$

$$PS + \frac{Pqr}{p + q + r} = QR + \frac{Qpr}{p + q + r} \quad (10)$$

$$QR = PS + \frac{Pqr - Qpr}{p + q + r} \quad (11)$$

Dividing by Q :

$$R = \frac{P}{Q}S + \frac{qr}{p + q + r} \left(\frac{P}{Q} - \frac{p}{q} \right) \quad (12)$$

If we design the bridge such that the ratio of the inner arms equals the ratio of the outer arms:

$$\frac{P}{Q} = \frac{p}{q} \quad (13)$$

Then the term $\left(\frac{P}{Q} - \frac{p}{q}\right)$ becomes zero. The balance equation simplifies to:

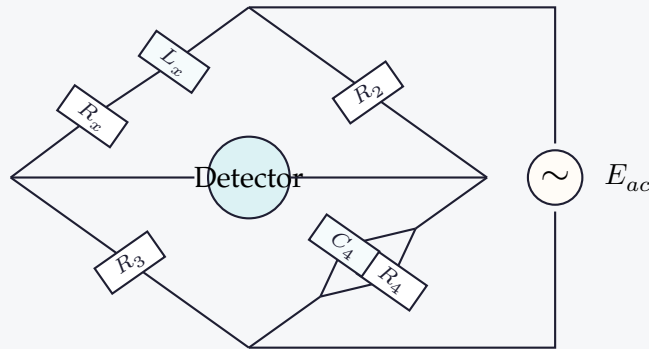
$$R = \frac{P}{Q}S \quad (14)$$

Thus, the effect of the connecting lead resistance r is completely eliminated.

1.3 Maxwell Inductance-Capacitance Bridge

The **Maxwell Bridge** is an AC bridge used to measure an unknown inductance by comparing it with a standard variable capacitor. It is suited for measuring medium-Q coils ($1 < Q < 10$).

Maxwell Inductance-Capacitance Bridge Circuit



1.3.1 Derivation of Balance Equation

Let:

- Arm 1 (AB): Impedance $Z_1 = R_x + j\omega L_x$ (unknown inductance and resistance).
- Arm 2 (AD): Impedance $Z_2 = R_2$ (non-inductive resistance).
- Arm 3 (BC): Impedance $Z_3 = R_3$ (non-inductive resistance).
- Arm 4 (DC): Impedance Z_4 consists of R_4 in parallel with C_4 . Thus, its admittance Y_4 is:

$$Y_4 = \frac{1}{R_4} + j\omega C_4 \quad (15)$$

The general AC bridge balance condition is:

$$Z_1 \cdot Z_4 = Z_2 \cdot Z_3 \implies Z_1 = \frac{Z_2 Z_3}{Z_4} = Z_2 Z_3 Y_4 \quad (16)$$

Substitute the values of Z_1 , Z_2 , Z_3 , and Y_4 :

$$R_x + j\omega L_x = R_2 R_3 \left(\frac{1}{R_4} + j\omega C_4 \right) \quad (17)$$

$$R_x + j\omega L_x = \frac{R_2 R_3}{R_4} + j\omega R_2 R_3 C_4 \quad (18)$$

Equating the real and imaginary parts on both sides:

$$\text{Real Part: } R_x = \frac{R_2 R_3}{R_4} \quad (19)$$

$$\text{Imaginary Part: } L_x = R_2 R_3 C_4 \quad (20)$$

The Quality Factor (Q) of the inductor is:

$$Q = \frac{\omega L_x}{R_x} = \frac{\omega(R_2 R_3 C_4)}{\left(\frac{R_2 R_3}{R_4}\right)} = \omega C_4 R_4 \quad (21)$$

1.3.2 Advantages and Disadvantages

- **Advantages:** Balance equations are independent of frequency; simple relation for computing L_x and R_x separately.
- **Disadvantages:** Requires an expensive variable capacitor; not suitable for high-Q coils ($Q > 10$) because the required resistance R_4 becomes extremely high, making it impractical.

2. Measurement of Non-Electrical Parameters

2.1 Flow Measurement: Electromagnetic Flow Meter

Electromagnetic flow meters measure the volumetric flow rate of an electrically conducting fluid.

Working Principle (Faraday's Law)

When a conductive fluid flows through a pipe of diameter D across a transverse magnetic field of flux density B , an electromotive force (EMF) E is induced across the electrodes perpendicular to the field:

$$E = B \cdot D \cdot v \quad (22)$$

where v is the average fluid velocity.

2.1.1 Concept and Working

- A magnetic coil placed around the pipe creates a uniform magnetic field.
- As the conductive fluid (measurand) flows, it acts as a moving conductor, generating voltage.
- Metallic electrodes flush-mounted on the pipe walls pick up the induced voltage.
- The flow rate Q is proportional to velocity: $Q = A \cdot v = \frac{\pi D^2}{4} \cdot \left(\frac{E}{BD}\right) = \frac{\pi D}{4B} \cdot E \implies Q \propto E$.

- **Advantages:** Non-invasive (zero pressure drop), handles corrosive/slurry liquids, independent of viscosity, density, and temperature. - **Disadvantages:** Fluid must be electrical-conducting ($> 5 \mu\text{S}/\text{cm}$); cannot measure oil, gas, or pure water.

2.2 Level Measurement

2.2.1 Capacitive Level Transducer

Operates on the variation of capacitance between two concentric cylindrical electrodes placed in a vessel. - **Non-Conducting Fluid:** Capacitance changes as the liquid level h rises, displacing air:

$$C = \frac{2\pi\epsilon_0[h\epsilon_r + (H - h)]}{\ln(b/a)} \quad (23)$$

where a and b are the inner and outer electrode radii, H is the total height, and ϵ_r is the liquid dielectric constant. - **Conducting Fluid:** The probe is insulated with a thin dielectric material (e.g., Teflon). The fluid acts as the outer plate of the capacitor. The capacitance varies linearly with depth h .

2.2.2 Ultrasonic Level Transducer

Non-contact type sensor mounted at the top of the tank. - **Working:** Emits ultrasonic pulses downward. The pulses reflect off the liquid surface and return to the transducer. - **Formula:**

The distance to the liquid surface d is calculated from transit time t :

$$d = \frac{v \cdot t}{2} \quad (\text{where } v = \text{velocity of sound in air}) \quad (24)$$

The liquid level is $h = H - d$, where H is the total height of the tank.

2.3 Pressure Measurement

2.3.1 LVDT-Based Pressure Cell

A mechanical pressure-sensing element (bellows, diaphragm, or Bourdon tube) is used as the primary transducer to convert pressure into physical displacement. The core of an LVDT is connected to this element as the secondary transducer, converting displacement into an electrical output voltage.

2.3.2 Pirani Vacuum Gauge

A thermal conductivity gauge used to measure low pressures (10^{-4} torr to 1 torr). - **Working:** A heated tungsten filament is placed in a chamber connected to the vacuum system. As pressure decreases, the density of gas molecules decreases. Fewer molecules collide with the filament, reducing heat dissipation (thermal conductivity drops). The filament temperature increases, raising its electrical resistance. This resistance change is measured using a Wheatstone bridge.

2.3.3 McLeod Vacuum Gauge

An absolute calibration instrument used to measure very low pressures down to 10^{-6} torr. - **Working:** A gas sample of large initial volume V at unknown low pressure P is compressed into a small capillary tube of area a to a much smaller volume. The compressed pressure is measured using a mercury column height difference h . - **Formula:** Applying Boyle's Law ($P_1 V_1 = P_2 V_2$):

$$P \cdot V = (P + h) \cdot (a \cdot h) \quad (25)$$

Since $P \ll h$ under compression, the unknown pressure is approximated as:

$$P \approx \frac{a \cdot h^2}{V} \quad (26)$$

3. Data Acquisition Systems (DAS)

Data Acquisition Systems interface physical sensors with computers to measure and process parameters automatically.

3.1 Analog and Digital DAS

- **Analog DAS:** Collects and displays data in continuous analog form (e.g., strip chart recorders, oscilloscope traces). - **Digital DAS:** Converts analog signals into digital values for computer storage, display, and processing.

Digital Data Acquisition System Block Diagram



3.2 Roles of Key Digital DAS Components

- **Analog Multiplexer (MUX):** Sequences through multiple sensor channels, connecting one signal at a time to the common measurement chain. This avoids replicating expensive S/H and ADC components for every channel.
- **Sample and Hold (S/H) Circuit:** Captures the rapidly changing analog voltage at a specific instant and holds it constant at its output. This prevents errors during the time the ADC takes to perform a conversion.
- **ADC (Analog-to-Digital Converter):** Converts the held analog voltage into a proportional binary number (digital word) for the computer.

Quick Revision — Module 5

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Key Module 5 Formulas

$$\text{Wheatstone: } R_x = \frac{Q}{P}R \quad \text{Kelvin Double: } R = \frac{P}{Q}S \left(\text{when } \frac{P}{Q} = \frac{p}{q} \right)$$

$$\text{Maxwell } L_x: L_x = R_2 R_3 C_4 \quad \text{Maxwell } R_x: R_x = \frac{R_2 R_3}{R_4}$$

$$\text{EM Flow EMF: } E = B \cdot D \cdot v \quad \text{McLeod Vacuum: } P \approx \frac{a \cdot h^2}{V}$$

One-Line Definitions

- **Kelvin Double Bridge:** A bridge with double ratio arms to eliminate contact and lead resistance errors.
- **Maxwell Bridge:** An AC bridge used to measure medium-Q coil inductance using standard capacitance.
- **Electromagnetic Flow Meter:** An inductive flow meter measuring conductive fluid speed via Faraday's Law.
- **Pirani Gauge:** A thermal conductivity vacuum gauge measuring pressure via filament heat loss.
- **McLeod Gauge:** A mercury-filled vacuum gauge that compresses low-pressure gas to measure it hydrostatically.
- **Multiplexer (MUX):** A switch routing multiple analog input channels sequentially to a single output line.
- **Sample and Hold (S/H):** A circuit holding analog voltage constant during the ADC conversion interval.

Frequently Asked / Viva Questions

Q1. Why does a standard Wheatstone bridge fail to measure low resistances accurately?

At low resistances ($< 1 \Omega$), the resistance of the connecting leads and contact terminals is significant relative to the test resistance. These lead resistances add directly to the measured resistance in a Wheatstone bridge, introducing massive measurement errors.

Q2. Show how the lead resistance is eliminated in a Kelvin double bridge.

By incorporating inner ratio arms p and q , the balance condition becomes $R = \frac{P}{Q}S + \frac{qr}{p+q+r}(\frac{P}{Q} - \frac{p}{q})$. Setting the outer ratios equal to the inner ratios ($\frac{P}{Q} = \frac{p}{q}$) cancels the second term, eliminating the lead link resistance r .

Frequently Asked / Viva Questions (contd.)

Q3. Derive the balance equations for a Maxwell inductance-capacitance bridge.

Setting the opposite arm impedances equal, we get $Z_1 = Z_2 Z_3 Y_4$. Expanding: $R_x + j\omega L_x = R_2 R_3 (\frac{1}{R_4} + j\omega C_4)$. Separating real and imaginary parts yields $R_x = \frac{R_2 R_3}{R_4}$ and $L_x = R_2 R_3 C_4$.

Q4. What are the limitations of a Maxwell bridge?

It is restricted to medium-Q coils ($1 < Q < 10$). For high-Q coils, R_4 must be extremely large, which is physically impractical. For low-Q coils ($Q < 1$), the balance converges very slowly due to sliding balance issues.

Q5. An AC Maxwell bridge has $R_2 = 1 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $C_4 = 0.1 \text{ }\mu\text{F}$, and $R_4 = 100 \text{ k}\Omega$. Find the unknown R_x and L_x .

$$\text{Given: } R_x = \frac{R_2 R_3}{R_4} = \frac{10^3 \times (2 \times 10^3)}{100 \times 10^3} = 20 \text{ }\Omega.$$

$$L_x = R_2 R_3 C_4 = 10^3 \times (2 \times 10^3) \times (0.1 \times 10^{-6}) = 0.2 \text{ H}.$$

Q6. Explain the working of an electromagnetic flow meter.

It operates on Faraday's Law ($E = BDv$). A magnetic field is applied across a pipe. The flowing conductive liquid acts as a conductor, generating an EMF across flush electrodes that is directly proportional to the average flow velocity.

Q7. What is the principle of a Pirani vacuum gauge?

It works on the principle that the thermal conductivity of a gas is proportional to its pressure at low pressure. As chamber vacuum increases, less heat is conducted away from the filament, raising its temperature and resistance.

Q8. Derive the pressure formula for a McLeod gauge.

Initial volume V at pressure P is compressed to volume $V_2 = a \cdot h$. The final pressure is $P + h$. By Boyle's Law: $PV = (P + h)ah = P ah + ah^2$. Since $P \ll h$, $PV \approx ah^2 \Rightarrow P \approx ah^2/V$.

Q9. Why is a Sample and Hold circuit necessary before the ADC in digital DAS?

An ADC requires a finite time to convert an analog voltage into a digital word. If the analog input changes during this conversion period, the ADC output will suffer from aperture error. The S/H holds the input constant during conversion.

Q10. Compare McLeod Gauge and Pirani Gauge.

McLeod gauge is an absolute, mechanical gauge working by volume compression, providing manual/discrete readings independent of gas type. Pirani gauge is a relative, thermal gauge providing continuous electrical output dependent on gas conductivity.

Q11. What is the role of an analog multiplexer in a multi-channel DAS?

It acts as a rotary switch to connect multiple analog inputs sequentially to a single shared S/H and ADC. This reduces cost by eliminating the need for separate ADCs for each channel.

McLeod Gauge vs Pirani Gauge Comparison

Feature	McLeod Gauge	Pirani Gauge
Principle	Gas volume compression (Boyle's Law).	Thermal conductivity variation.
Reading Mode	Discrete/Manual (batch process).	Continuous electrical reading.
Gas Dependency	Independent of gas composition.	Highly dependent on gas type.
Output Signal	Hydrostatic column level (height h).	Electrical resistance change.
Calibration	Absolute gauge (no calibration needed).	Relative gauge (needs calibration).
Range	10^{-6} torr to 10^{-2} torr	10^{-4} torr to 1 torr